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Roll No

CE-226-CBCS

B.E., IV Semester

Examination, June 2020

Choice Based Credit System (CBCS)

Water Supply and Waste Water Engineering - I

Time : Three Hours

Maximum Marks : 60

- Note:** i) Attempt any five questions
ii) All questions carry equal marks.

1. a) Enumerate the various surface sources of water and discuss and compare the quality and quantity of water supplies that may be available from these sources.
b) Give a reasonable estimate of average daily consumption of water for a residential town of 50,000 population. Suggest a possible break-up of consumption on different heads.
2. Explain the following water treatment process mentioning impurity removed by them
 - i) Aeration
 - ii) Screening
 - iii) Plain settling
 - iv) Filtration
 - v) Disinfection
 - vi) Zeolite process
 - vii) Settling plus coagulation
3. a) Write a detailed note on the detection and prevention of wastage of water in the distribution system.

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- b) State the functions of service Reservoir and sketch the sectional elevation of the same, showing the various appurtenances.
- 4. a) What are the various types of storm water regulators used in a sewerage system. Explain briefly working of one of these with sketch.
b) When pumping of sewage is required in a sewerage system? How is it more difficult than the pumping of water in water supply schemes? Draw a sketch of a typical pumping station.
- 5. Explain the procedure of estimating BOD of a given sample of sewage. The five day BOD at 20°C of a given sample is 380 mg/L. Calculate the ultimate BOD at 38°C given the deoxygenation constant at 20°C as 0.1 per day.
- 6. a) Explain the factors affecting self purification property of a stream?
b) Differentiate between dilution method and land disposal method for sewage disposal.
- 7. a) Write a note on water borne diseases and their control.
b) Given the following data, calculate the population at the end of next three decades.

Year	1971	1981	1991	2001
Population	80,000	1,20,000	1,68,000	2,28,580

- 8. Write short note on:
 - a) Hardy cross method
 - b) Per capita demand
 - c) pH value of water
 - d) Oxygen sag analysis

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